

Course outline

Course Code: 1929		
Name of the Programme: Skill Enhancement		
Name of the Course: Structured Innovation with Design Thinking		
Semester: I		
Course credit	No. of hours per week	Total no. of Teaching hours
2	1:0:2	45
Course Objectives:		
• Apply design thinking for creative problem-solving • Synthesize concepts for sustainable prototype development • Enhance existing products focusing on market aesthetics • Innovate within frameworks to create new prototypes and communicate		

Syllabus:

Module - 1 Design thinking for creative problem-solving	No. of hours
	10
Introduction to Design Thinking: Overview of design thinking principles and stages. Empathy in Design: Techniques for understanding and empathizing with users, including user interviews and persona creation. Storytelling as a Tool: Using storytelling to convey user needs and problem statements. Case Studies: Analyzing successful design thinking projects that started with empathy. Workshops: Hands-on activities to practice empathic engagement with hypothetical user scenarios.	
Module - 2 Concepts for sustainable prototype development	No. of hours
	10
Basics of Recycling and Upcycling: Definitions, differences, and environmental impact. Triple Constraints in Product Design: Exploring how time, scope, and budget affect product development. Prototype Development: Step-by-step guidance on creating prototypes using upcycled materials. Sustainability in Design: Discussing the importance of sustainable practices in modern design. Practical Exercises: Designing and building simple products using recycled materials within defined constraints.	
Module - 3 Existing products focusing on market aesthetics	No. of hours
	10
Understanding Product Aesthetics: How aesthetics influence consumer behavior and product design. Detailed Product Analysis: Dismantling and evaluating existing products to understand their design and functionality. Enhancement Techniques: Methods for improving product design, focusing on aesthetics and functionality. Consumer Feedback: Gathering and utilizing consumer feedback to guide redesign efforts. Reconstruction Projects: Students reconstruct an existing product with improved aesthetics and functionality.	

Module - 4 Frameworks to create new prototypes	No. of hours
	10
Innovation vs. Invention: Exploring the difference and focusing on innovation within existing frameworks. Analyzing Market Trends: How to use market research to identify opportunities for innovation. Creative Adaptations: Techniques for creative thinking and adaptation of existing products. Prototyping with Purpose: Building prototypes that address specific user needs or market gaps. Case Studies and Discussions: Reviewing examples of successful innovations in existing product lines.	

Module - 5 Communicate project processes and outcomes	No. of hours
	5
The Art of Storytelling: Techniques for crafting compelling narratives around product development. Presentation Skills: Tips and strategies for effective public speaking and presentation. Visual and Digital Tools: Using visual aids, digital presentations, and storytelling software to enhance communication. Engaging Stakeholders: How to pitch and explain projects to different audiences (investors, customers, peers). Project Documentation and Feedback: Methods for documenting the development process and incorporating feedback for improvement.	

Lab Plan	[30 Hours]
1	Day 1: Empathy Activity & Paper Prototyping
2	Day 2: Define & Ideate Activity
3	Day 2: Upcycling
4	Day 3: Product study
5	Day 3: Product variation & Testing
6	Day 4: Product development: Game Design for a purpose

Course outcomes
CO1 Apply design thinking for creative problem-solving
CO2 Synthesize concepts for sustainable prototype development
CO3 Enhance existing products focusing on market aesthetics
CO4 Innovate within frameworks to create new prototypes and communicate

Text books

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| 1. | Jake Knapp, John Zeratsky, and Braden Kowitz, "Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days" Simon & Schuster (2016), ISBN: 978-1501121746, 1st ed. |
| 2. | Idris Mootee "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School" Wiley (2013), ISBN: 978-1118620120, 1st ed. |

Reference books

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| 1. | Jeanne Liedtka and Tim Ogilvie "Designing for Growth: A Design Thinking Toolkit for Managers" Columbia University Press (2011) ISBN: 978-0231158381, 1st ed. |
| 2. | Pavan Soni, Design Your Thinking, Penguin Random House India Private Limited (2020), 978-0670094097, 1st ed. |